

**A  
Project Report  
On  
Enteropathy-Associated T-Cell Lymphoma  
in Celiac Disease: A Comprehensive  
Epidemiological and Clinical Analysis  
Submitted in partial fulfilment of the requirements  
for the award of the degree of  
Bachelor of Technology  
In  
Information Technology  
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**GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY**

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### **DECLARATION**

We hereby declare that the project work presented in this project report entitled **“Enteropathy-Associated T-cell Lymphoma in Celiac Disease : A Comprehensive Epidemiological and Clinical Analysis ”** in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Information Technology, submitted to A.P.J.Abdul Kalam Technical University, Lucknow, is based on my work carried out at Department of Information Technology, Galgotias college of engineering and technology, Greater Noida. The work contained in the report is original and project work reported in this report has not been submitted by me/us for the award of any other degree or diploma.

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## **CERTIFICATE**

This is to certify that the project report entitled “**Enteropathy-Associated T-cell Lymphoma in Celiac Disease : A Comprehensive Epidemiological and Clinical Analysis**” submitted by **Aditya Kumar Tiwari (2200970130007), Ayush Yadav (2200970130038), Harshit Somvanshi (2200970130049), Hritik Tiwari (2200970130051)** to the AKTU, Uttar Pradesh in partial fulfilment for the award of Degree of Bachelor of Technology in Information Technology is a Bonafide record of the project work carried out by them under my supervision during the year 2024-2025.

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## ABSTRACT

Enteropathy-associated T-cell lymphoma (EATL) is an uncommon and aggressive malignancy of the gastrointestinal tract that arises from intraepithelial T-cell lymphocytes and is closely associated with Celiac disease, particularly type II refractory celiac disease. The disease leads to severe malabsorption and rapidly spreads within the small intestine, often resulting in life-threatening complications such as intestinal perforation and hemorrhage, which significantly limit therapeutic options. The latest World Health Organization (WHO) 5th edition classification of haematolymphoid tumours clearly distinguishes EATL from monomorphic epitheliotropic intestinal T-cell lymphoma, emphasizing differences in pathogenesis, morphology, and genetic alterations.

This paper provides a comprehensive overview of EATL, incorporating epidemiological data on its incidence and survival trends derived from population-based studies. Furthermore, it explores key molecular mechanisms, including dysregulation of interleukin-15 signaling, alterations in transforming growth factor-beta pathways, and initiating mutations in the JAK–STAT signaling cascade. Advanced diagnostic techniques, such as high-throughput T-cell receptor sequencing, are also discussed for their role in assessing clonal diversity. The limitations of conventional anthracycline-based chemotherapy are highlighted, and emerging therapeutic strategies are reviewed, including targeted therapies such as CD30-directed agents like Brentuximab vedotin. In addition, ethical considerations related to genetic testing and clinical study designs are addressed. Overall, this paper aims to provide a refined and integrated understanding of EATL to improve its diagnosis and management.

**KEYWORDS:** *Celiac disease, Enteropathy-associated T-cell lymphoma (EATL), intraepithelial lymphocytes, refractory celiac disease, JAK–STAT signaling pathway, targeted therapy*

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# **CHAPTER 1**

## **INTRODUCTION**

EATL, or Enteropathy Associated T-cell Lymphoma is an aggressive cancer of the gastrointestinal tract and has the potential to be fatal [1]. It is classified as an extralymphatic (extranodal) mature T-cell, non-Hodgkin lymphoma that primarily invades the mucosal and submucosal layers of the small intestine [5]. While the gastrointestinal tract is the most common site for extranodal lymphomas, the majority of extranodal lymphomas arise from B-cells; thus, there are few primary intestinal T-cell lymphomas. In fact, EATL accounts for 13% - 15% of all gastrointestinal lymphomas and <1% of all NHL globally [1,5].

The specialized immune cells responsible for EATL are intraepithelial lymphocytes, which are found within the epithelial lining of the intestine, where they act as the first point of immune defence. Under physiological conditions, the function of intraepithelial lymphocytes is to protect the intestine from infection and disease; however, in certain pathological conditions, intraepithelial lymphocytes may develop into cancerous cells. Chronic inflammation, particularly in patients with chronic Celiac Disease, is a major contributor to the malignant transformation of intraepithelial lymphocytes [4,6]. Chronic immune stimulation and mucosal injury lead to a pro-oncogenic environment that provides the genetic instability needed for the development of lymphoma [6].

EATL has a rapid progression with multiple foci of small intestine involvement and the following complications: intestinal perforations, obstructions, and GI bleed [2,3]. Presenting symptoms are often nonspecific (mild symptoms) and include abdominal pain, chronic diarrhea, weight loss, and malabsorption, all of which cause a delay in diagnosis until the disease is in an advanced stage [2].

Patients with EATL have a poor prognosis with a median survival time of 7–10 months and a 5-year survival rate of <20% [1,2]. Delays in diagnosis, aggressive biological behaviour of EATL, and malnutrition all have an impact on the poor prognosis associated with the disease. The combined impact of severe malnutrition, hypoalbuminemia and cachexia decreases the ability of patients to tolerate aggressive

chemotherapy and adversely affects treatment outcomes [3].

Our knowledge of how EATL develops has changed dramatically due to advances in molecular biology. Evidence suggests that disruption of the JAK-STAT pathway contributes to abnormal proliferation and/or longevity of T-cells (1). In addition, new studies on epidemiology, survival differences, and clinical outcomes are contributing to better classification and management strategies for EATL (1,2,5). New therapies are also being developed, which could lead to better patient outcomes (5).

### **1.1 Background and Motivation**

Enteropathy associated T-cell lymphoma is one of the absolute most aggressive; biologically complex; and lethal types of malignancy, found in the human gastrointestinal tract [1]. This malignant neoplasm is classified as an extranodal mature T-cell, non-Hodgkin lymphoma; it primarily affects but is not limited to the mucosa and submucosa of the small intestine [2]. While the gastrointestinal tract is the most common anatomical region in which extranodal lymphomas present; most are expected to be B-cell derived; primary intestinal T-cell lymphomas, however, are exceptionally rare in clinical practice, accounting for 13–15% of all gastrointestinal lymphomas and, consistently, < 1% of the total burden of global non-Hodgkin lymphoma.<sup>1</sup> Enteropathy associated T-cell lymphoma arises from a distinct sub-population of highly specialized immune cells called intraepithelial lymphocytes. These cells are found throughout the epithelial lining of the intestinal mucosa, forming the first line of immunological defence against pathogens [1].

Enteropathy-associated T-cell Lymphoma is characterized by a close and profound pathophysiological connection to celiac disease, a systemic autoimmune disorder that occurs in genetically predisposed individuals after the consumption of gluten protein [1].

The intestinal mucosal immune system normally has a highly regulated and balanced tolerance to commensal microbiota and to dietary antigens.

When individuals with HLA-DQ2 or HLA-DQ8 (over 90% of diagnosed cases) are presented with deamidated gluten peptides via antigen-presenting cells, they trigger a localized chronic inflammatory response that causes local immune regulation to break down over a period of decades from the constant antigenic stimulation and mucosal

inflammation.

In some patients with refractory celiac disease, clinical characteristics include the presence of severe malabsorption symptoms ("symptoms"). Persistent diarrhea and severe villous atrophy generally persists for more than 12 months despite strict compliance to eating a gluten-free diet [1].

Refractory celiac disease can be divided into two categories: Type I and Type II. Type I continues to have a polyclonal and phenotypically normal population of intraepithelial lymphocytes that are similar to active celiac disease and Type II is characterized by the presence of an abnormal monoclonal population of intraepithelial lymphocytes that display very poor phenotype [2]. The very aberrant Type II intraepithelial lymphocytes are usually negative for expression of CD3 surface antigen and the CD8 co-receptor, but they express strong intracellular CD3 and have clonal rearrangements of T-cell receptor (TCR) gamma chains [1]. Therefore, hematopathologists describe Type II refractory celiac disease as a low-grade T-cell lymphoma (intraepithelial TCL) or essentially an in situ malignant process [1]. If not treated promptly and effectively, the indolent clonal expansion will acquire additional genetic lesions and progress to a rapidly growing and aggressive high-grade neoplasm classified as enteropathy-associated T-cell lymphoma [1].

## **1.2 Problem Statement**

Enteropathy-associated T-cell lymphoma remains an extremely difficult-to-manage hematologic malignancy with a very poor prognosis even after 20 years of great progress in the areas of genomic profiling, sub-classification, and targeted biologic therapy in treating hematologic malignancies. Evidence of the severe challenge posed by this malignancy is the overwhelming number of patients who die from it each year in every part of the world; the median overall survival is only approximately 7-10 months from initial diagnosis and the cumulative 5-year overall survival rates have remained at less than 20% for decades with little change [1].

The main problem regarding the diagnosis of celiac disease stems from the gradual nature of this disease and the resulting significant delay before both an accurate diagnosis is made and an appropriate treatment is initiated. Patients with early stages of lymphoma frequently present with clinical symptoms that do not specifically point towards celiac disease such as unrelieved diarrhea, generalized upset stomach, and

progressive malnutrition / cachexia. Because the cancerous lesions are typically located near the inside of the folds (plica) that line the jejunum or proximal ileum of the intestine and because of anatomical complexities associated with these areas, the malignant lesions do not generally have a clear signature when evaluated with either standard upper endoscopy or lower colonoscopy procedures. Because of this, a high percentage of celiac disease patients (up to 50%) are ultimately diagnosed emergently after presenting to an emergency department with one or more serious, potentially fatal, abdominal complications (spontaneous intestinal perforation, acute bowel obstruction, or massive gastrointestinal hemorrhage).

The patient's biological status of nutrition and physiological function is often very limited (whether it be severely low) by the time they receive a firm diagnosis or have begun receiving therapy. Malnutrition and nutritional deficiencies (i.e. severe hypoalbuminemia – typically  $<21.6$  g/L) due to malabsorption related to the enteropathy play a significant role in the decline of this status [10] with dramatic impacts on the patient's ability to tolerate aggressive curative intent chemotherapy administered using multiple agents ([6]).? In addition, the cancer cells that exist in the patient exhibit intrinsic resistance to most standard chemotherapy treatments like anthracyclines, which results in high levels of refractoriness to initial treatment and very high rates of post-treatment mortality due to relapse of disease [1]. The intersection of late diagnosis, emergent surgical needs, increased cachexia, and treatment resistance, together creates an urgent need for new diagnostic markers, new nutritional interventions and new target/therapeutics architectures.

### **1.3 Objective**

This extensive research report provides a thorough examination of the major gaps in clinical and diagnostic information described in the problem statement, through meeting three main objectives. First, this analysis attempts to compile the global incidence and demographic information available on this disease by quantifying the amount of the population affected by it using large population-based registries. In addition, the report will identify differences in disease incidence rates between different geographic regions and geographic areas. Additionally, the report will provide a summary of the survival rates based on geographic area. Second, the report seeks to comprehensively describe the disease's many complex mechanisms at the

cellular level by mapping the biological pathways involved in the progression of this malignancy, including the for example the pathology of the gluten-interleukin-15 signaling axis and the involvement of chromosomal aberrations. Lastly, this report will attempt to clarify the current changes in the definition of the disease that the World Health Organization has put forth, and to provide distinguishing criteria between classic forms of the disease and monomorphic epitheliotropic intestinal T-cell lymphoma, so that pathologists can use these to accurately identify patients in each group based on histology, immunophenotype and other genetic factors associated with these two types of malignancies.

## **CHAPTER 2**

### **LITERATURE REVIEW**

Several researchers have studied the relationship between gluten sensitivity and risk for cancer due to the system-wide implications of gluten-related disorders over the last several years. This section provides a summary of the important contributions made to the scientific literature regarding the findings, strengths, and weaknesses of the research conducted thus far regarding the complexities associated with these two concepts.

**Sajjad Bakhtiari et al. (2024)** study examined the association between consumption of gluten and occurrence of gastrointestinal cancers among persons suffering from gluten-related disorders such as celiac disease (CD). Untreated patients with CD could be at an increased risk for malignant disease including enteropathy-associated T-cell lymphoma (EATL) and small bowel adenocarcinoma due to their continuous exposure to gluten. The study results collectively suggest that chronic exposure to gluten contributes to the development of malignant disease because of the effects of chronic inflammation and immune dysregulation on cellular processes that promote malignant disease development. Ultimately, however, the researchers concluded that there is insufficient information to assess any cancer risk in persons with non-T-cell immunodeficiency or other forms of non-celiac gluten sensitivity (NCGS), and further longitudinal studies should be conducted to evaluate these populations for cancer risk.

**Gastroenterology and Hepatology from Bed to Bench (2024)** paper examined cancer related risks attributed to individuals with coeliac disease (CD) based on molecular mechanisms involved in the development of cancer in CD patients. The study found that cytokine overproduction and increased intestinal permeability related to the immune system are the two primary mechanisms contributing to development of cancer from tumours (i.e., the formation of new cells or cell clusters derived from original cell types). While the work is extremely informative regarding risks associated with CD, there was limited evidence linking non-celiac gluten sensitivity (NCGS) with similar risk due to the absence of autoimmune markers. Therefore, the authors recommend developing more sensitive diagnostic tools for use in assessing

NCGS-related cancer risks.

Shewry et al. (2018) assessed how gluten composition can affect human health, particularly the protein content and immunogenicity of glutenins within gluten. They further illustrate the role of agricultural practices used to improve nitrogen fertilizer applications on gliadin levels and glutenin levels, further increasing the associated potential immunogenicity of wheat proteins. Therefore, even though the researchers demonstrate how agricultural practices can influence gluten sensitivity, they do not produce conclusive evidence linking these agricultural practices to cancer risk, creating an opportunity for further research.

Vazquez-Roque et al. (2021) verified whether a GFD could eliminate inflammation and lower cancer occurrence in patients with CD. Results revealed that strict GFD adherence greatly reduced the risk of developing cancer, especially gastrointestinal cancers. However, participants in the GFD diet also had lower concentrations of certain nutrients like dietary fibres and vitamins A and D, which could lead to a higher probability of developing cancer. The authors suggested the use of dietary supplements and/or balanced alternative gluten-free products to rectify such deficiencies.

Ziegler et al. (2020) reported on the effect of gut microbiota on the relationship between gluten and the development of cancer. Dysbiosis (reduction of beneficial gut flora and an increased prevalence of disease causing flora) is affecting those with CD, causing high levels of inflammation and creating a carcinogenic environment. Although this study has provided new insights into changes in gut flora, larger studies will be needed to confirm the existence of a causative relationship between change of GI flora and those with NCGS.

Chen et al. (2022) looked at NCWS cancer-related long-term risk. The researchers felt this low-grade inflammation and increased gut permeability from NCWS could lead to higher cancer risk. They concluded there is a lack of valid research to identify long-term implications for NCWS due to unacceptable standards for diagnosing this disorder.

Penuelas et al. examined how farmers adding nitrogen to soils in contemporary agriculture is part of the increases associated with gluten sensitivity and resulting from



gluten sensitivity and cancer. They feel that nitrogen-enriched wheat gives greater levels of gliadins (a type of immunogenic protein) — gluten proteins — this will increase the amount of gluten sensitive individuals will react to through immune systems. The authors show strong evidence for a relationship between agricultural practices and gluten sensitivity and offer further evidence for how environmental factors may contribute to gluten sensitivity-related cancer risk [6].

Wang et al. conducted gluten consumption and its relation to cancers of the digestive tract showed that high levels of gluten are associated with an increased risk of proximal colon cancer among men but did not show any increased risk overall for colorectal cancers. Their work identified the need for more studies on how gluten plays a role in the development of cancer among individuals with gluten intolerance [5].

Elfström et al. found that patients with celiac disease and either inflammation or latent celiac disease had a lower likelihood of developing GI cancers if they were on a gluten-free diet. However, they also noted an increased risk for lymphoma in untreated celiac patients which indicates that early diagnosis and dietary modification are c

Taraghikhah et al. describe their update to the gluten-related disorders, highlight any previous research that may have focused only on clinical or diagnostic features of the condition, and look at the continued variability in symptoms and difficulties experienced by clinicians attempting to differentiate between coeliac disease (CD), non-gluten coeliac syndrome (NCGS), and wheat allergy. The need for additional biomarker development and improved diagnostic testing to aid in diagnosis and management are emphasized in their article [12].

## **2.1 Evolution of Clinical and Molecular Understanding**

Over the last few decades the study and classification of enteropathy-associated T-cell lymphoma has changed considerably on a conceptual and clinical level. Enteropathy-associated T-cell lymphoma was first described in the medical literature during the mid-1900s as "malignant histiocytosis of the intestine," and it was incorrectly assumed that the disease was derived from the macrophage-histiocyte lineage because of the massive inflammatory infiltrate, presence of many eosinophils, and greatly variable appearance of the cellularity under light microscopy [1]. It was not

until landmark immunohistochemical studies were performed during the late 1980s that the malignant proliferation was found to have a T-cell immunophenotype, confirming this condition's placement among the various other peripheral T-cell lymphomas [1]. The subsequent identification of a specific mucosal homing integrin (now known as CD103) that is expressed at a high level on the malignant cells provided definitive proof that these malignancies derive from the highly specialized intraepithelial T-lymphocyte population. [1]

The systematic connection between long-term, potentially undetected or ineffectively treated celiac disease and the subsequent occurrence of aggressive lymphoma has been formulated through exhaustive retrospective cohort studies [1]. A landmark study by Malamut et al. provided an accurate forecast of the survival rates for individuals diagnosed with celiac disease based solely on the type of enteropathy and the nutritional status at the time of diagnosis.<sup>1</sup> The results of this study showed an obvious two-tier prognosis: the five-year survival rate for individuals who developed lymphoma from a background of non-clonal celiac disease and type I refractory celiac disease was nearly 58.8 percent, while the five-year survival rate for those who developed lymphoma directly from monoclonal type II refractory celiac disease was precisely zero percent [1]. This landmark investigation also established two independent significant predictors of improved overall survival, namely: (i) baseline serum albumin greater than 21.6 grams per liter, and (ii) the incorporation of primary debulking surgery prior to the initiation of systemic chemotherapy [1].

Through the use of high-throughput sequencing techniques that profile adaptations of the T-cell Receptor (TCR) repertoire, researchers were able to establish a very clear and strong correlation of TCR loss in patients with celiac disease vs patients with overt lymphoma. Scientists have established a clear correlation between the natural history of celiac disease and the aforementioned changes in TCR repertoire. A large body of evidence exists to support the assertion that; as celiac disease crosses into the type II refractory stage (an advanced stage of the disease), the highly diverse and polyclonal TCR repertoire collapses completely and is replaced by multiple and unregulated expansions of malignant-accessible TCR clonotypes that are malignant and private to each patient. Using state-of-the-art sequencing technologies, researchers have demonstrated that the dominant malignant TCR clonotypes share no sequence

homology with publicly available gluten-reactive TCRs and confirm the currently accepted view that the malignant transformation from an antigen-mediated process (i.e., gluten-mediated) to autonomous malignancy has occurred.

## **2.2 Epidemiological Trends and Global Disparities**

Due to the extreme infrequency of celiac-associated lymphedema (CAL) and historical inaccuracies with clinical documentation as a severe form of celiac relapse, precise epidemiological tracking of this particular cancer has been hindered. However, comprehensive data analysis related to CAL will become available beginning in 2023 through the use of the National Cancer Institute's (NCI) vast population-based Surveillance, Epidemiology and End Results (SEER) program database [2]. A recent two-decade retrospective CAL incidence analysis (2000-2021) performed through use of the SEER program database identified a precise, age-standardized annual incidence for CAL of 0.014/100,000 population [1]. Though CAL remains extremely infrequent overall, longitudinal data provide strong evidence to suggest an alarming trajectory, with CAL incidence gradually increasing at an annualized percentage change of 2.63% [2]. Based on this empirical evidence, longitudinal CAL incidence increases have a significant positive correlation with increases in the global number of cases of celiac disease and improvements in high-resolution endoscopic and molecular diagnostic techniques that accurately characterize CAL.

The geographical distribution of the celiac disease is characterized by great variability with regard to the demographic characteristics of the populations that are affected and the genetic characteristics associated with the population genetics of the respective HLA-DQ2 and HLA-DQ8 susceptibility alleles within the same geographic areas.[1] Examples can be found in Northern European countries with a high rate of celiac disease, where 9.1 percent of peripheral T-cell lymphomas diagnosed in patients with Enteropathy will have an associated or causative role in the development of enteropathy-associated T-cell lymphomas. A slightly lower rate of 5.8 percent is recorded in the United States and Canada for the same disease, whereas in Asian countries where the celiac-associated HLA haplotypes are present at a very low frequency, enteropathy-associated T-cell lymphomas will comprise only 1.9 percent of peripheral T-cell lymphomas.[1] Furthermore, the majority of intestinal T-cell lymphomas arising from within these Asian populations are monomorphic epithelial

specific intestinal T-cell lymphomas which were previously misclassified as a unique subtype and do occur de novo and are not etiopathogenically or genetically related to any form of gluten enteropathy.[1]

### **2.3 Therapeutic Milestones and Chemotherapeutic Trials**

Academic writing related to various treatment options for an illness typically contains mostly retrospective cohort studies as well as multiple smaller size, multi-country prospective studies. This is primarily because the statistical odds of doing an adequately powered large-scale randomized controlled study of an orphan disease are virtually nil [2]. The International Peripheral T-Cell Lymphoma Project has provided scientific information about the clinical treatment of over 1,300 individuals from different areas throughout the world with T-cell lymphomas and confirmed that peripheral T-cell lymphoma is an entirely different, aggressive type of lymphoma from other types in relation to treatment and management needs [1].

Traditionally, physicians have used standard CHOP regimens (cyclophosphamide, doxorubicin, vincristine and prednisone) as their mainstay for treating patients with intestinal non-Hodgkin lymphoma. According to current literature, research has shown definitively that CHOP is not only not optimal treatment but also is inferior to other new therapies as demonstrated in a landmark clinical trial performed by the German Studygroup on Intestinal Non-Hodgkin Lymphoma, who found that although there may be some small benefit seen from using CHOP in localized early stage cases, it does not confer any substantial survival benefit to the vast majority of patients presenting at an advanced disseminated stage.

Through the joint collaborative effort of Scottish and Newcastle, there was a major, paradigm-changing milestone achieved for therapeutic benefit by the application of high-dose, intensive induction chemotherapy followed by autologous stem cell rescue, which has previously not been attempted. The study utilized an alternating, highly aggressive regimen of ifosfamide, etoposide, and epirubicin in conjunction with an intermediate-dose of methotrexate and culminated with an autologous stem cell transplant after myeloablative BEAM conditioning [3]. This intensive regimen led to substantially improved clinical outcomes, with the greatest reported complete response rate of 72 percent at 5 years and an overall survival rate of nearly 67 percent

at 5 years for patients who were physiologically fit to undergo and complete the intense regimen, thereby establishing a new benchmark for transplant-eligible patients [1].

The significant advancement made in understanding the relationship between gluten sensitivity and cancer risk (especially in CD patients) noted in this literature review is offset by the identification of several gaps (e.g., the need for longitudinal studies on NCGS and better diagnostic tools). Correcting these limitations will be fundamental to improving the ability to refine dietary interventions and positively impact the outcomes of people with gluten-related disorders. Future studies should focus on exploring the molecular mechanisms of the inflammation, immune modulation, and gut permeability involved in cancer pathogenesis.

Furthermore, the assessment of modern agricultural practices explicating greater immunogenicity of gluten should be evaluated to reduce health risks associated with consumption of wheat.

There is mounting evidence now establishing that patients with celiac disease have a defined risk of developing malignancies due to chronic inflammation and dysregulation of the immune system, while non-celiac gluten sensitivity presents a risk that is less clearly defined but could plausibly be substantiated by further studies. The combination of dietary interventions with advancing testing methods and changes in agricultural practices to prevent cancers in gluten-sensitive individuals may ultimately provide a better way to reduce the occurrence of cancer eligibility in these groups.

This literature review documents important advances related to our knowledge regarding the relationship between gluten sensitivity and risk of cancer. On one hand, patients with celiac disease are at a proven increased risk of developing malignancies due to both chronic inflammation and immune dysregulation; however, non-celiac gluten sensitivity (NCGS) patients still have an unclear association with development of malignancy from gluten (the exact methods remain to be identified). Although, this literature review documents substantial strides towards the understanding of gluten sensitivity as it relates to cancer risk, there remain significant limitations, including; 1) lack of longitudinal studies in patients with NCGS, 2) lack of reliable diagnostic methodologies. It is important to address these limitations to continue the refinement

of dietary interventions as well as ensuring that individuals with gluten-related disorders achieve optimal outcomes.

| <b>Pa<br/>pe<br/>r<br/>N<br/>o.</b> | <b>Year</b> | <b>Research<br/>Name</b>                                               | <b>Resear<br/>ch<br/>Metho<br/>dology</b>        | <b>Statisti<br/>cal<br/>Metho<br/>ds</b>              | <b>Key Findings</b>                                                                                                                                                                                                                        | <b>Gaps<br/>Identified</b>                                                                                                | <b>Scope for<br/>Future<br/>Research</b>                                                                                                         |
|-------------------------------------|-------------|------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>                            | 2021        | Immune-mediated Diseases and Cancer Risk (UK Biobank cohort)           | Prospective cohort (UK Biobank)                  | Multivariable Cox proportional hazards (time-varying) | Slightly increased total cancer risk with immune diseases (HR $\approx$ 1.08); celiac disease had high risk (HR $\approx$ 6.9) for small intestine                                                                                         | Rare outcomes (celiac and small bowel cancer are uncommon); short follow-up for rare cancers                              | Larger cohorts or meta-analyses for rare cancer outcomes; study mechanisms linking autoimmunity to cancer                                        |
| <b>2</b>                            | 2022        | Cancer Risk in 47,241 with Celiac (nationwide Swedish cohort)          | Population-based matched cohort (record linkage) | Stratified Cox regression for hazard ratios           | Overall slightly increased cancer risk (HR $\approx$ 1.11) in celiac patients vs controls; excess risk limited to first year after diagnosis (HR $\approx$ 2.47 in year 1, HR $\approx$ 1.01 thereafter)                                   | Lack of data on diet compliance or gluten-free diet adherence; limited to Swedish population                              | Examine effect of strict gluten-free diet on cancer risk; similar studies in other populations; analyze specific cancer types over time          |
| <b>3</b>                            | 2017        | Long-term Gluten Consumption and Coronary Heart Disease (Nurses' HPFS) | Prospective cohort (NHS I/II, HPFS)              | Cox proportional hazards                              | No meaningful correlation exists between gluten intake and coronary heart disease (CHD) after multivariate adjustments. When comparing the highest and lowest quintiles of consumers based on the gluten effect on CHD (HR $\approx$ 0.95; | Observational design; dietary gluten intake correlated with whole/refined grain intake (confounding); only US middle-aged | Further investigation of specific grain components (refined vs whole grain); randomized trials of gluten restriction vs not (unlikely), focus on |

|   |      |                                                                |                                                      |                              |                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                              |                                                                                                                                                                                  |
|---|------|----------------------------------------------------------------|------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |      |                                                                |                                                      |                              | 95% CI = 0.88 - 1.02), there was no significant difference. Interestingly, when gluten intake was evaluated alone without considering whole grains, it appeared to correlate inversely with the incidence of CHD.                                                                                                                                                | health professionals (mostly white)                                                                                                          | whole-grain intake health effects                                                                                                                                                |
| 4 | 2022 | Gluten Intake and Digestive Cancers (3 US cohorts)             | Prospective cohort (NHS I/II, HPFS)                  | Multivariable Cox regression | When evaluating the association of dietary gluten to risk of colorectal, gastric, esophageal, liver or pancreatic cancer after multivariate adjustment, no statistically significant correlation was found (HR $\approx$ 0.94; Ptrend $\approx$ 0.05). As such, there was insufficient evidence to suggest a significant impact on individual digestive cancers. | Excluded individuals with celiac disease; gluten intake estimated from diet (measurement error); relatively short follow-up for some cancers | Studies in other populations; examine potential interactions (e.g. high-gluten diet and gut microbiome); randomized trials not feasible, but Mendelian randomization if possible |
| 5 | 2020 | Celiac Disease and Small Intestinal Tumors (Nationwide Sweden) | Population-based cohort (pathology registry linkage) | Stratified Cox regression    | Celiac patients had significantly elevated risk of small-bowel adenocarcinoma (HR $\approx$ 3.05) and adenomas (HR $\approx$ 5.73) vs controls; no increased risk of small-bowel carcinoids (HR $\approx$ 0.59). Absolute risk is low (1 extra adenocarcinoma per $\sim$ 3000 patient-years over 10 yrs).                                                        | Rare outcomes require very large cohorts or meta-analyses; no data on diet compliance or latency period                                      | More research on screening strategies for high-risk CD patients; molecular studies on pathogenesis of small-bowel cancers in CD                                                  |
| 6 | 2020 | Celiac Disease and                                             | Population-based                                     | Stratified Cox               | Celiac disease (CD) was found to be associated with a modest increase in                                                                                                                                                                                                                                                                                         | Unmeasured confounders (e.g.                                                                                                                 | Investigate long-term impact of strict                                                                                                                                           |

|   |      |                                                                 |                                        |                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                  |                                                                                                                                                                              |
|---|------|-----------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |      | Mortality (Swedish population)                                  | matched cohort (register linkage)      | regression                                   | all-cause mortality when compared to the general population (HR $\approx$ 1.21). Significant elevations were seen in deaths from cardiovascular disease (CVD; HR = 1.08), cancer (HR $\approx$ 1.29), and respiratory causes (HR $\approx$ 1.21) were found. Participants with a diagnosis of celiac disease showed the highest risk of mortality, at 1 year post-diagnosis (HR $\approx$ 2.34)                                                                                                   | comorbidities); lack of data on diet adherence or disease severity                                                               | gluten-free diet on mortality; explore interventions to reduce specific risks (e.g. cancer surveillance, cardiovascular prevention)                                          |
| 7 | 2022 | Nutrient Intake in Celiac and Gluten-Related Disorders (NHANES) | Cross-sectional analysis (NHANES data) | Comparative statistics (t-tests, regression) | Diagnosed celiac disease (CD) patients (n = 14) had lower energy and nutrient intakes than controls (n = 14). Diagnosed celiac patients were shown to have a greater intake of calories, fibre and micronutrients as compared to undiagnosed celiac. The non-celiac gluten avoiders (PWGs) had reduced intakes of carbohydrates and sugars. Nutritional imbalances among groups may play a role in the difference in incidence of GI cancers between diabetes and gluten-free diet (GFD) groups.. | Cross-sectional design; self-reported diet; cannot infer causality; unknown if nutrient differences translate to health outcomes | Longitudinal studies to track nutrition over time; assess health outcomes (e.g. bone health, CVD risk) of observed deficiencies; guidance on GFD nutritional supplementation |



|   |      |                                                      |                                |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                        |                                                                                                                                                                      |
|---|------|------------------------------------------------------|--------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 | 2022 | Celiac Disease and Cancer: Latest Evidence (Review)  | Narrative review of literature | Not applicable (review, no new stats) | Celiac disease (CD) and gluten-related disorders (GRDs) not receiving medical treatment are associated with increased risks for developing GI malignancies (colorectal and lymphomas), whereas gluten consumption appears to have a protective effect regarding certain malignancies to non-CD individuals. It can be concluded that avoidance of gluten in CD will prevent certain types of cancer, while gluten may provide a protective benefit in the non-CD population. | Broad statements without new data; potential bias in literature selection; need quantitative synthesis | More meta-analyses and cohort studies to quantify protective effects of gluten vs risk in GRDs; mechanistic studies on gluten's role in carcinogenesis or prevention |
| 9 | 2021 | Wheat/Gluten Disorders & GFD Misconceptions (Review) | Narrative review               | Not applicable (review)               | The authors review the classification of gluten-related disorders (Celiac Disease [CD], Non-Celiac Gluten Sensitivity [NCGS], and wheat allergy) and myths related to a gluten-free diet (GFD). They point out that a GFD is critical for CD and will help relieve many symptoms from other conditions such as NCGS, while also mentioning that when someone adopts a GFD for inappropriate reasons, they will have nutritional deficiencies,                                | Not primary research; does not quantify risks; general discussion                                      | Calls for public education and more research on GFD effects in non-CD individuals; need RCTs or cohorts on long-term GFD outcomes in different populations           |

|    |      |                                                             |                                            |                                              |                                                                                                                                                                                                                                                           |                                                                                                           |                                                                                                                                                                                               |
|----|------|-------------------------------------------------------------|--------------------------------------------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      |                                                             |                                            |                                              | and the GFD may not lower disease risk for individuals who are healthy.                                                                                                                                                                                   |                                                                                                           |                                                                                                                                                                                               |
| 10 | 2024 | Gluten and Cancer: PPI Network Analysis (Review)            | Review with computational network analysis | Protein–protein interaction network analysis | Concludes gluten elimination prevents inflammation in GRDs and thus may prevent associated cancers; notes gluten consumption in non-GRDs might be linked to lower cancer risk. Highlights lack of strong evidence for gluten as cancer prevention factor. | Narrative with no quantitative data; speculative on protective effect; PPI analysis not validated in vivo | Experimental research on gluten’s bioactive effects; prospective studies on gluten intake and cancer incidence; refine targets from PPI analysis                                              |
| 11 | 2023 | Celiac Disease and Cardiovascular Risk (Case–Control)       | Retrospective case–control (J Clin Med)    | Logistic regression (multivariate)           | Celiac disease (with long-term GFD) was associated with *lower* risk of cardiovascular disease, especially carotid plaques (11.8% vs 40.1% in controls. Authors conclude CD (on GFD) reduces CVD risk.                                                    | Retrospective design; potential selection bias; unclear if GFD or CD drives results; single-center        | Prospective studies on GFD duration and CVD outcomes; investigate contradictory findings (vs other studies showing higher CVD risk); mechanistic studies on GFD impact on lipids/inflammation |
| 12 | 2022 | Gluten Intake and Inflammatory Bowel Disease (3 US cohorts) | Prospective cohorts (NHS I/II, HPFS)       | Multivariable Cox regression                 | Among 5+ million person-years, higher gluten intake (highest vs lowest quintile) was not associated with Crohn’s disease (HR≈1.16, p=0.41) or ulcerative                                                                                                  | Dietary gluten estimated by questionnaire ; cohort mostly middle-aged                                     | Explore in diverse populations; investigate other dietary factors or genetic                                                                                                                  |

|    |      |                                                       |                                              |                                       |                                                                                                                                                                                                                                         |                                                                                                                                 |                                                                                                                                                                 |
|----|------|-------------------------------------------------------|----------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      |                                                       |                                              |                                       | colitis (HR≈1.04, p=0.64) incidence, Gluten consumption did not increase IBD risk.                                                                                                                                                      | health professionals (not general pop); observational                                                                           | susceptibilities; mechanistic studies on gluten and gut immunity                                                                                                |
| 13 | 2019 | Gluten Intake and Microscopic Colitis (US women)      | Prospective cohorts (NHS I/II women)         | Multivariable Cox regression          | No association between long-term gluten intake and microscopic colitis risk (220 cases in ~20 years). Adjusted HRs ~1.00 for highest vs lowest gluten intake (ptrend=0.88), Null effect held across subtypes (collagenous/lymphocytic). | Only included women (mostly Caucasian nurses); gluten intake self-reported; incidence of microscopic colitis is low (219 cases) | Similar studies in men or other populations; examine specific diet components (refined vs whole grains); assess role of other food sensitivities (e.g. lactose) |
| 14 | 2021 | Gluten Intake and Cognitive Function (NHS)            | Prospective cohort (NHS women)               | Multivariable linear regression       | Long-term gluten intake was *not* associated with global or domain-specific cognitive scores in middle-aged women without CD. No support for restricting gluten to preserve cognition in absence of disease                             | Only middle-aged women; short cognitive follow-up (4 yrs apart); diet assessed repeatedly but cognitive testing started later   | Studies in older or male cohorts; longer-term cognitive follow-up; interventional studies of GFD on cognitive function; neuroimaging studies in CD vs controls  |
| 15 | 2024 | Celiac Disease and Mortality (Systematic Review/Meta) | Systematic review/meta-analysis (25 studies) | Pooled hazard ratios (random-effects) | Meta-analysis found CD patients have modestly increased all-cause mortality (HR≈1.16) and malignancy-related mortality (HR≈1.21), driven largely by markedly elevated non-Hodgkin lymphoma mortality (HR≈10.14).                        | High heterogeneity (I <sup>2</sup> up to 96%); variability by region and decade; mix of diagnosed vs undiagnosed,               | Better-quality prospective data; focus on cause-specific mortality in well-characterized cohorts; impact of early diagnosis and strict GFD on                   |

|    |      |                                                            |                                                |                                              | Mortality trends improved in recent decades                                                                                                                                                                                                                                                                              | treated vs untreated                                                                                                     | long-term survival                                                                                                                                                |
|----|------|------------------------------------------------------------|------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 | 2023 | Nutrient Intakes on Gluten-Free Diet (SR/Meta)             | Systematic review and meta-analysis            | Random-effects meta-analysis                 | Gluten-free diet (in CD patients) tended to be high in fat and low in essential micronutrients (iron, B vitamins, etc.). GFD patients had average 48% carbs, 15% protein, 37% fat, with low fiber and micronutrients. Authors caution that long-term GFD may predispose to CVD or bone issues if nutrients not monitored | Only dietary intake data (no health outcomes); mostly single-arm (no control group); publication bias possible           | Long-term studies on health outcomes (fractures, CVD events) in GFD CD patients; interventions to improve GFD nutritional quality; guidelines for supplementation |
| 17 | 2015 | Celiac Disease and Bone Fractures (Systematic Review/Meta) | Systematic review/meta-analysis (obs. studies) | Pooled relative risks (fixed/random-effects) | CD was associated with ~30% increased risk of any fracture and ~69% increased hip fracture risk in prospective studies. In cross-sectional/case-control analysis, fractures were ~2× more common in CD patients                                                                                                          | Majority of studies are older; possible confounding by DEXA screening bias; uncertain role of gluten-free diet adherence | More prospective data on fracture risk in treated vs untreated CD; evaluate bone mineral density improvement on GFD; assess benefits of vitamin D/calcium in CD   |
| 18 | 2018 | Celiac Disease and Colorectal Adenomas (Systematic)        | Systematic review/meta-analysis (3 studies)    | Pooled odds ratios (fixed-effect)            | No significant difference in colorectal adenoma prevalence: CD patients vs controls OR≈0.94 (95% CI 0.65–1.38) for any adenoma; OR≈0.97 (0.48–1.97) for advanced adenomas, Conclusion:                                                                                                                                   | Only 3 studies (367 CD, 682 controls); no analysis of colorectal cancer; heterogeneity                                   | Larger endoscopy-based studies; examine CRC incidence (not just adenomas) in CD; account for surveillance                                                         |

|    |      |                                                                   |                                                                   |                                                              |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                             |                                                                                                                                                                                                                                  |
|----|------|-------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      | Review/<br>Meta)                                                  |                                                                   |                                                              | CD not linked to higher<br>colonic adenoma risk.                                                                                                                                                                                                                                                                                                                                  | low but<br>evidence<br>sparse                                                                                                               | bias (CD<br>patients may<br>have more<br>colonoscopies)                                                                                                                                                                          |
| 19 | 2016 | Celiac<br>Disease<br>and<br>Overall<br>Cancer<br>Risk<br>(Meta)   | System<br>atic<br>review/<br>meta-a<br>nalysis<br>(17<br>studies) | Pooled<br>ORs<br>(fixed/r<br>andom<br>-effects<br>)          | CD modestly increased<br>pooled risk of all cancers<br>(OR≈1.25) and GI<br>cancers (OR≈1.60).<br>Elevated risk mainly in<br>peri-diagnosis period<br>(OR≈1.72) but not after<br>(OR≈1.09), suggesting<br>delayed diagnosis and<br>role of GFD. Small<br>absolute risk (~1%).                                                                                                      | Study<br>heterogeneity<br>; older<br>literature;<br>varied case<br>definitions;<br>GFD<br>adherence<br>not assessed                         | Prospective<br>cohort studies<br>on cancer<br>incidence in<br>CD; detailed<br>analysis of<br>gluten<br>exposure and<br>timing;<br>effectiveness of<br>GFD in<br>reducing<br>cancer risk                                          |
| 20 | 2024 | Celiac<br>Disease<br>and<br>Fertility<br>(Review)                 | System<br>atic<br>review<br>(narrati<br>ve)                       | Not<br>applica<br>ble<br>(revie<br>w)                        | Celiac Disease has been<br>linked to fertility<br>problems and adverse<br>pregnancy outcomes<br>(miscarriage, preterm<br>birth). There is some<br>evidence to suggest that a<br>gluten-free diet may help<br>improve fertility but is<br>very limited. The<br>authors' request is for<br>additional studies<br>examining the impacts of<br>a gluten-free diet and<br>infertility. | Mostly small<br>studies and<br>surveys;<br>heterogeneity<br>in study<br>designs;<br>confounding<br>(malnutrition<br>vs<br>autoimmunit<br>y) | Large cohort<br>studies on<br>fertility and<br>pregnancy in<br>diagnosed vs<br>undiagnosed<br>CD; trials of<br>GFD in<br>infertility of<br>unknown<br>cause;<br>mechanistic<br>studies on<br>autoimmunity<br>and<br>reproduction |
| 21 | 2024 | Celiac<br>Disease<br>and Mood<br>Disorders<br>(Meta-ana<br>lysis) | System<br>atic<br>review/<br>meta-a<br>nalysis                    | Pooled<br>odds<br>ratios<br>from<br>psychi<br>atry<br>scales | Patients diagnosed with<br>CD have been shown to<br>have a much higher<br>chance of developing<br>anxiety and depression<br>(OR=2.26 and OR=3.36,<br>respectively) than                                                                                                                                                                                                           | Heterogeneo<br>us<br>assessment<br>tools;<br>cross-section<br>al data;<br>confounding                                                       | Longitudinal<br>studies on<br>mood<br>before/after<br>GFD; elucidate<br>mechanisms<br>(e.g.                                                                                                                                      |

|    |      |                                                                                               |                                       |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |                                                                                                                                                                                                                    |
|----|------|-----------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      |                                                                                               |                                       |                                             | controls. The GFD improved the anxiety and depression scores of Celiac Disease patients over the course of one year.                                                                                                                                                                                                                                                                                                                                   | by symptom burden                                                                                         | inflammation, nutrient malabsorption); mental health screening in CD clinics                                                                                                                                       |
| 22 | 2020 | Gluten Intake and Metabolic Health (UK Biobank)                                               | Cross-sectional analysis (UK Biobank) | Multiple linear regressions with adjustment | There are weak associations between gluten intake and metabolic factors; higher gluten intake is associated with a lower BMI for men but a greater BMI for women, as well as smaller differences in the level of lipid measurements. The overall associations, though statistically significant, have very little clinical significance. The conclusion is that gluten restriction will most likely not positively impact metabolic health in general. | Cross-sectional design (no causality); 24-h recall diet estimate; UK population (mostly Caucasian)        | Longer-term cohort analyses; clarify gender differences; examine whole-grain vs refined grain contributions; RCTs of gluten restriction not feasible, but dietary pattern interventions could test this indirectly |
| 23 | 2018 | Gluten Intake and Type 2 Diabetes Risk (3 US cohorts): contentReference[oaicite:44]{index=44} | Prospective cohorts (NHS I/II, HPFS)  | Cox proportional hazards                    | Higher gluten intake has been *inversely* associated with Type 2 Diabetes risk (to ~12 g/day); the highest levels of intake had a lower incidence of T2D (linear inverse relationship): contentReference[oaicite:45]{index=45}. Probably the result of increased intake of                                                                                                                                                                             | Observational; strong correlation of gluten with other carbs and fiber; mostly white health professionals | Investigate specific grain components vs diabetes risk; randomized dietary fiber interventions to clarify causation; generalize to other ethnicities/countries                                                     |

|    |      |                                                           |                                                     |                                           |                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |                                                                                                                                                      |
|----|------|-----------------------------------------------------------|-----------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |      |                                                           |                                                     |                                           | whole grain (fibre), not due to gluten by itself.                                                                                                                                                                                                                                                                                                                      |                                                                                                                                            |                                                                                                                                                      |
| 24 | 2014 | Celiac Disease and Pregnancy Complications (Survey study) | Retrospective survey-based cohort (tertiary center) | Descriptive stats, chi-square comparisons | Women with celiac disease had a 50.6% spontaneous abortion rate and a 23.6% rate of premature birth compared with women without celiac disease, with rates of 40.6% and 15.9%, respectively. Menarche occurred at a later age on average among women with celiac disease. Fewer women with celiac disease reached reproductive age than did women without the disease. | Recall bias (online survey); selection bias (tertiary center and volunteers); no adjustment for other risks                                | Prospective studies on pregnancy outcomes in screened CD; impact of early diagnosis and GFD on fertility/pregnancy; population-based linkage studies |
| 25 | 2020 | Gluten Intake and Mortality (UK Biobank)                  | Prospective cohort (UK Biobank)                     | Cox regression (cause-specific mortality) | No statistically significant association exists between gluten intake and mortality for all diseases or from cancer (HR $\approx$ 1.00); however, gluten intake was positively correlated with death from ischaemic heart disease. Therefore, limiting one's gluten content through a gluten free diet may not enhance one's longevity.                                | Mostly healthy middle-aged adults; residual confounding possible; small effect sizes may be chance; gluten intake assessed by 24-h recalls | Replicate in other cohorts; explore mechanism of IHD finding; longer follow-up; analyse specific diet patterns beyond gluten alone                   |
| 26 | 2023 | Celiac Disease and Cardiovas                              | Narrative review of                                 | Not applicable                            | It has been suggested in recent studies that patients with celiac disease are at increased                                                                                                                                                                                                                                                                             | Heterogeneous findings; conflicting data (some                                                                                             | Prospective global studies on CVD incidence in                                                                                                       |

|  |  |                                       |                                    |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                     |                                                                                                                           |
|--|--|---------------------------------------|------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|  |  | cular<br>Disease<br>Risk<br>(Review): | epidem<br>iology<br>literatu<br>re | (review) | <p>risk of cardiovascular disease (including ischaemic heart disease, myocardial infarction and atrial fibrillation) despite having fewer conventional risk factors. The affect of the gluten free diet on cardiovascular disease is incompletely understood, with some studies showing lower cholesterol levels whilst on the diet. Consequently, increasing the focus of celiac disease prevention of cardiovascular disease is indicated.</p> | <p>show lower CVD); most studies in Western populations; few long-term GFD data</p> | <p>CD; trials of diet/exercise tailored to CD; study how GFD composition (often high fat) affects cardiovascular risk</p> |
|--|--|---------------------------------------|------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

Table 2.1 : Literature Survey



## CHAPTER 3

### DESIGN AND METHODOLOGY

#### 3.1 Study Design and Implementation Scope

The implemented methodology for the CeliGuard ML system has been documented in this chapter as they appear in the current repository artifacts. The CeliGuard ML product development adhered to a standard Machine Learning workflow that divides the process of developing a model, serving predictions from that model, and interacting with users into independent services. This approach enables increased reproducibility, simplified debugging, and the ability to control the update of a model without changing its public prediction interface.

This chapter is designed to clearly make the difference between the conceptual intent of the project and the measurable implementation of that project. Many of the user facing documents (and notes) within this repository use the term malignancy risk, however, the actual supervised celiac\_serology\_risk target is described within the training notebook and/or serialised metadata. Since any methodology needs to be reproducible and able to be audited, it is imperative to document in this chapter what is trained/validated/deployed/returned from any code paths—the data provided is not implicitly implied from terminology used in prior to this document (e.g. malignancy risk).

The system's architecture contains three executable parts, which are the Training Pipeline, the FastAPI Inference Backend, and the Streamlit Frontend. These three components communicate only via stable file outputs and HTTP contracts. The Model Artifacts created by the Training Pipeline are saved in a directory called "models." The FastAPI Inference Backend (backend) reads the Model Artifacts at startup to create RESTful API endpoints. The Streamlit Frontend (frontend) interacts with each of these backend endpoints to create an interactive experience for users to assess risk.

##### 3.1.1 Component Responsibilities

The Training Service handles Data Ingestion, Data Preparation, Model Comparison, Model Decision Weight Tuning, Model Selection, and exportation of all Model and

Metadata artifacts.

The Backend Service performs validation of request payloads against a strict schema, executes model inference operations, and maps numeric classes to their respective Label names; therefore, returning probabilities as well as an explanation of each class. Lastly, the Frontend Service is responsible for acquiring User inputs using bounded (limited) forms, checking backend availability (for each user), submitting Prediction Requests to the backend, and rendering class-level risk and recommendations blocks back to the User.

Additionally, there is a Shared Model Location (on storage) for backward compatibility for all Training Outputs to their respective Model Outputs during the Inference Runtime, as indicated by `celiac_risk_model.pkl` and `model_metadata.pkl`.

### **3.1.2 Reproducibility Principles**

The implementation provides for reproducibility due to explicit dependencies, deterministic random states of model training, and fixed schemas of all input variables. Wherever possible, the same feature names will be reused between preprocessing steps in the notebooks, within model fields in the API request, and as labels for controls in the front end. This decreases the likelihood of schema drift occurring. Additionally, outside of the code, runtime configuration has been externalized using environment variables (i.e., `MODEL_PATH`, `METADATA_PATH`, `API_URL`) so that the same code can run in both local and containerized environments.

Finally, isolation of the service also allows for repeatable deployments. The back-end and front-end processes run continuously, and training employees will only run once on demand (as a batch job). This prevents any accidental retraining that can occur during inference run-time and makes it easier to keep track of artifact lineage.

### **3.2 Data Acquisition Strategy**

The executed guidance/handbook for training notebook uses data (information) from Hugging Face using their datasets API. Data source information is maintained in the metadata and currently points to `"hf:nhanes/nhanes_celiac_data/train"` and `"hf:nhanes/nhanes_celiac_data/test"`. When loading the training and test splits, the source descriptor information is also added to the resulting combined metadata from

the notebook for downstream research and access.

This approach decouples source retrieval from subsequent model logic; therefore, should the upstream dataset change over time, the same notebook code can be executed again to provide comparable preprocessing and evaluation steps, however, the ultimate behavior of the resulting models will depend upon the stability of the source schema and class distributions in the retrieved splits.

### **3.2.1 Target Definition and Label Space**

The training target has been set to `celiac_serology_risk`. The effective label space provided for the target represents an ordinal style encoding consisting of 3 classes (0=Low, 1=Moderate, 2=High). Target preprocessing coerces the values for the target to integer format, impute all invalid or missing entries as class 1, and clips the target labels to the inclusive range of 0-2. This method of defining target labels will help to prevent interruption of training from noisy target labels while also possibly helping to improve the ability of the model to handle very rare examples of target data points that were defined according to target classification by the degree of noise possible from other types of training provided to the model.

### **3.2.2 Feature Schema**

The system uses 13 clinical-style predictors. Five are numeric and eight are categorical. Numeric features are `age_at_diagnosis`, `current_age`, `years_of_symptoms_before_diagnosis`, `bmi`, and `followup_years`. Categorical features are `sex`, `marsh_grade_at_diagnosis`, `mucosal_healing_on_followup`, `rcd_type`, `smoking_status`, `gfd_adherence`, `family_history_of_malignancy`, and `hla_risk`. The same fields are enforced in backend request validation and surfaced in frontend inputs.

## **3.3 Data Preprocessing Methodology**

Preprocessing is implemented as a two-stage procedure: fit-time state estimation on training data and deterministic application on each split. During state fitting, per feature numeric medians are computed for imputation and per-feature categorical fallback values are computed from valid in-domain modes. The code checks for required columns before any transformation and fails early when mandatory schema

elements are absent.

When applying preprocessing, numeric values are converted with coercion, missingness is detected, and values are clipped to predefined valid intervals. For each numeric feature, a corresponding missing-indicator field is generated. Categorical values are stripped and validated against allowed category sets; invalid values are replaced with training-derived fallback categories. Rows are then filtered for completeness across model-required columns.

1. `age_at_diagnosis`: 5 to 80
2. `current_age`: 5 to 90
3. `years_of_symptoms_before_diagnosis`: 0 to 15
4. `bmi`: 16 to 35
5. `followup_years`: 0 to 20

Although missing-indicator columns are generated in preprocessing logic, metadata for the current exported model records `active_missing_indicators` as empty. This means the selected feature set for the deployed artifact includes only base numeric features and categorical features. The presence of indicator-generation code still provides future extensibility if retraining activates those columns.

### **3.4 Modeling Framework and Candidate Strategies**

Model development compares two classifier families implemented through scikit-learn Pipelines: Logistic Regression and Random Forest. Both models share a common `ColumnTransformer` preprocessing stage containing `StandardScaler` for numeric columns and `OneHotEncoder(handle_unknown="ignore")` for categorical columns. Shared preprocessing ensures candidate comparisons are driven by classifier behavior rather than inconsistent feature engineering.

For Random Forest, the implementation uses `n_estimators=200`, `max_depth=10`, fixed `random_state`, and `n_jobs=-1`. Logistic Regression uses `max_iter=1000` with optional class-weight balancing. Candidate balancing strategies include `none`, `class_weight_only`, `oversample_only`, and `class_weight_plus_oversample`. Oversampling is implemented through classwise upsampling to the largest class count within each training fold.

### 3.5 Selection Criteria and Exported Artifact

Model selection is performed by lexicographic ranking using cross-validated weighted minority recall mean as the first key, then weighted macro-F1, weighted balanced accuracy, and holdout accuracy. An epsilon tie window of 0.002 is applied to identify near-equivalent candidates, after which simpler strategies are preferred when performance is comparable. Strategy complexity prioritization favors minimal balancing interventions when possible.

The selected artifact, according to `models/model_metadata.pkl`, is Random Forest with strategy `class_weight_only` and decision weights `{0: 1.3, 1: 1.0, 2: 1.2}`. Exported metadata contains numeric and categorical feature lists, strategy flags, evaluation metrics, source descriptor, target descriptors, and explicit limitations text. This metadata is loaded by backend endpoints and should be treated as the canonical run summary for deployed behavior.

- Holdout accuracy: 0.9272
- Holdout balanced accuracy: 0.8720
- Holdout macro F1: 0.9093
- Holdout minority recall mean: 0.8102
- Cross-validation accuracy: 0.9382
- Cross-validation balanced accuracy: 0.8893
- Cross-validation macro F1: 0.9230
- Cross-validation weighted macro F1: 0.9211
- Cross-validation weighted balanced accuracy: 0.8889
- Cross-validation weighted minority recall mean: 0.8360
- Cross-validation folds: 5

### 3.6 Inference API Methodology

The FastAPI backend initializes model and metadata during application lifespan startup. File paths are configurable through environment variables, with defaults pointing to shared models directory artifacts. If artifact loading fails, the service remains process-alive but reports unavailable model state in health and prediction flows, enabling operational diagnostics without immediate process crash.

The predict endpoint accepts validated input through a Pydantic model with explicit

numeric bounds and field descriptions. Input payloads are converted into a single-row pandas DataFrame to match the model pipeline interface. The endpoint calls `predict` and `predict_proba`, maps numeric class outputs to text labels, and attaches a rule-based clinical-style explanatory message generated from salient input factors.

The response contract contains `risk_class`, `risk_score` vector, and `message`. This contract is stable for frontend consumption and can be versioned in future releases if additional metadata is required. Backend also exposes `/model-info` to retrieve model type, accuracy, and feature families from metadata for transparency and audit support.

### **3.7 Frontend Interaction Methodology**

The Streamlit frontend enforces structured data collection via typed controls: number inputs for numeric variables and select boxes for categorical variables. Numeric fields follow the same bounded ranges as backend validation to reduce client-server mismatch. Before prediction, the frontend performs backend health checks and stops interaction flow if API connectivity fails.

After successful prediction, the frontend displays class-specific styling, probability metrics for all three classes, explanatory text from backend response, and recommendation templates keyed on predicted class. The interface thereby functions as a presentation layer and intentionally delegates all model computation to the backend service.

## CHAPTER 4

### IMPLEMENTATION

#### 4.1 Introduction

CeliGuard ML is implemented as a complete machine-learning application with three tightly integrated but operationally separate layers: model training, model serving, and user interaction. The implementation is repository-first, meaning deployment behavior follows code and artifact contracts rather than abstract architecture descriptions. The repository layout reflects this separation directly:

- `train/` contains the executed notebook pipeline and dependencies required to build model artifacts.
- `backend/` contains the FastAPI inference service that loads the trained model and exposes prediction APIs.
- `frontend/` contains the Streamlit UI that collects patient-style inputs and renders prediction outputs.
- `models/` contains serialized model artifacts and generated model-report charts used for interpretability and reporting.

The goal of this documentation is to present the actual implementation and the rationale for each component choice, including the exact request/response contracts, preprocessing behavior, model-selection strategy, and deployment wiring. This file intentionally emphasizes verifiable behavior from source code and saved artifacts over generalized ML narratives.

There is one critical implementation-level caveat that should be explicit from the beginning. Although some historical project wording references malignancy risk, the currently saved training metadata indicates:

- `target_column = celiac_serology_risk`
- `not_malignancy_model = True`

That means the model currently deployed and served by backend endpoints should be interpreted as a celiac serology risk classifier with three classes (Low, Moderate, High), not as a clinically validated malignancy diagnosis model. The implementation

itself is consistent with that metadata.

The implementation uses service separation for reliability and maintainability:

1. **Training service** runs as a batch pipeline and writes model artifacts.
2. **Backend service** starts as a long-running API daemon and serves inference.
3. **Frontend service** runs as a long-running web app and consumes backend APIs.
  - This separation is intentional for three reasons:
  - It keeps heavy training dependencies and workflows out of serving runtime.
  - It allows model retraining and backend redeployment independently.
  - It keeps frontend logic thin and deterministic by delegating all model decisions to backend.

At runtime, the operational data flow is:

- 1. User enters values in Streamlit form.
- 2. Frontend builds JSON payload and calls backend /predict.
- 3. Backend validates payload, converts to DataFrame, invokes loaded pipeline.
- 4. Backend returns class label, probability vector, and explanation text.
- 5. Frontend renders class card, probabilities, interpretation, and recommendations.

The interface contract between backend and frontend is stable and concise:

- Request schema: 13 required fields.
- Response schema: risk\_class, risk\_score (3 values), message.

Because this contract is explicit in code, schema drift is easy to detect and fix during development.

## 4.2 Implementation of Training Pipeline

It is notebook-based but functionally structured, with helper functions for:

- data loading
- preprocessing state fitting
- split preprocessing



- model training and evaluation
- balancing strategy comparison
- decision-weight tuning
- artifact serialization

The pipeline is reproducible because dependencies are pinned and random-state usage is explicit in model and cross-validation routines.

The target field is:

- celiac\_serology\_risk

Class space is:

- 0 -> Low
- 1 -> Moderate
- 2 -> High

This mapping is preserved in backend inference code and frontend display logic.

#### **4.1.1 Feature Schema**

The model uses 13 features:

##### **Numeric (5):**

- age\_at\_diagnosis
- current\_age
- years\_of\_symptoms\_before\_diagnosis
- bmi
- followup\_years

##### **Categorical (8):**

- sex
- marsh\_grade\_at\_diagnosis
- mucosal\_healing\_on\_followup
- rcd\_type
- smoking\_status
- gfd\_adherence

- family\_history\_of\_malignancy
- hla\_risk

This schema is implemented consistently across training constants, backend request schema, and frontend form fields.

#### 4.1.2 Preprocessing Logic

Preprocessing is implemented as a two-stage strategy:

1. **fit preprocessing state on train split**
2. **apply preprocessing state to train/test splits**

Core behavior:

- Required-column validation before processing.
- Numeric coercion and clipping.
- Numeric median imputation.
- Missing-indicator generation for numeric fields.
- Categorical domain validation and fallback replacement.
- Target coercion, fill, and clipping.

Configured numeric ranges are:

- age\_at\_diagnosis: 5-80
- current\_age: 5-90
- years\_of\_symptoms\_before\_diagnosis: 0-15
- bmi: 16-35
- followup\_years: 0-20

The pipeline defines missing-indicator fields, but metadata from the current exported artifact shows:

- active\_missing\_indicators = []

So currently selected deployed features are base numeric + categorical sets.

#### 4.3 Candidate Models and Strategy Comparison

Implemented model families:

- Logistic Regression
- Random Forest

Both are wrapped in a shared preprocessing ColumnTransformer:

- StandardScaler for numeric features
- OneHotEncoder(handle\_unknown="ignore") for categorical features

Balancing strategies evaluated:

- none
- class\_weight\_only
- oversample\_only
- class\_weight\_plus\_oversample

Cross-validation:

- Stratified 5-fold
- per-fold and aggregated metrics captured

Primary optimization direction:

- minority recall mean (classes 0 and 2)

Supporting metrics:

- macro F1
- balanced accuracy
- overall accuracy

#### **4.4 Decision Weight Tuning**

The pipeline includes post-probability decision calibration by searching class weights (especially for boundary classes) on out-of-fold probabilities. This is a practical, implementation-level mechanism to improve minority sensitivity without changing model family or rewriting objective functions.

Current metadata reports:

- model\_type: Random Forest
- selection\_strategy: class\_weight\_only

- decision\_weights: {0: 1.3, 1: 1.0, 2: 1.2}

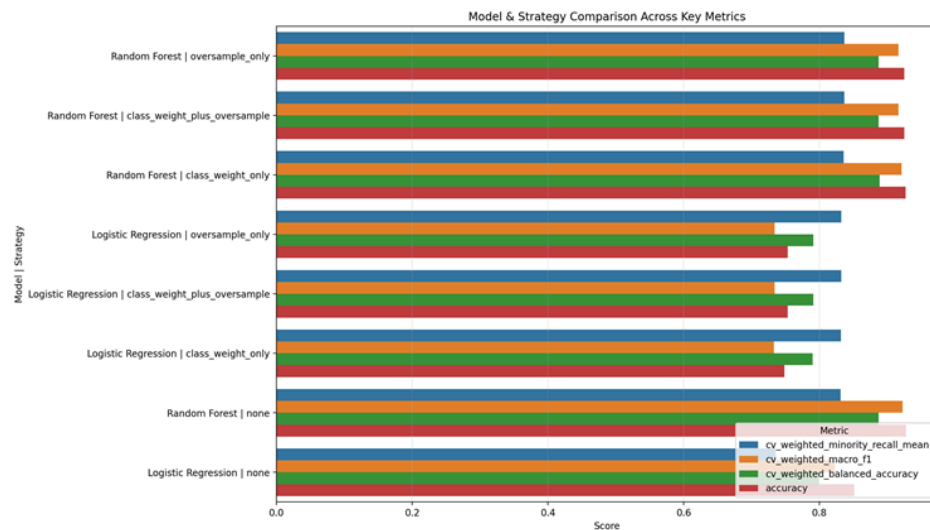
Core performance values:

- accuracy: 0.9272007310386842
- balanced\_accuracy: 0.8719969286791951
- macro\_f1: 0.9092636239644786
- minority\_recall\_mean: 0.8101703567693967
- cv\_weighted\_minority\_recall\_mean: 0.8360412927931483

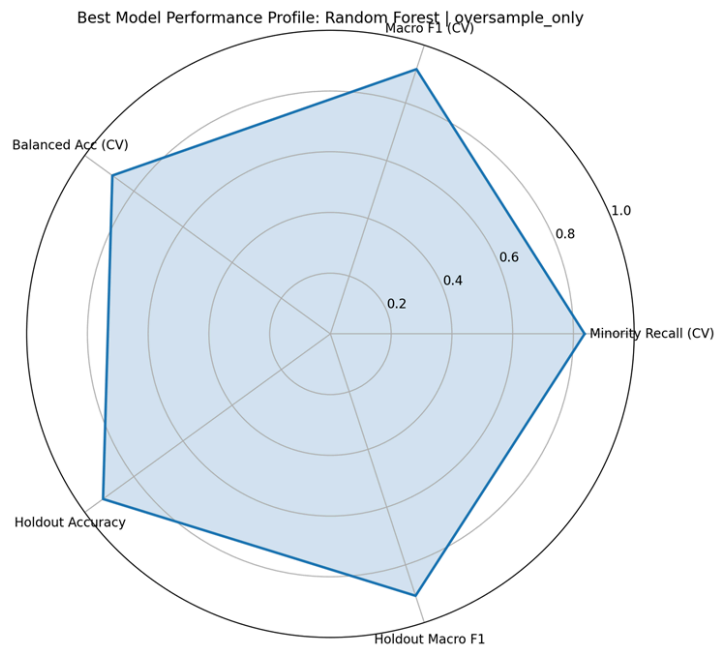
Artifacts saved:

- models/celiac\_risk\_model.pkl
- models/model\_metadata.pkl

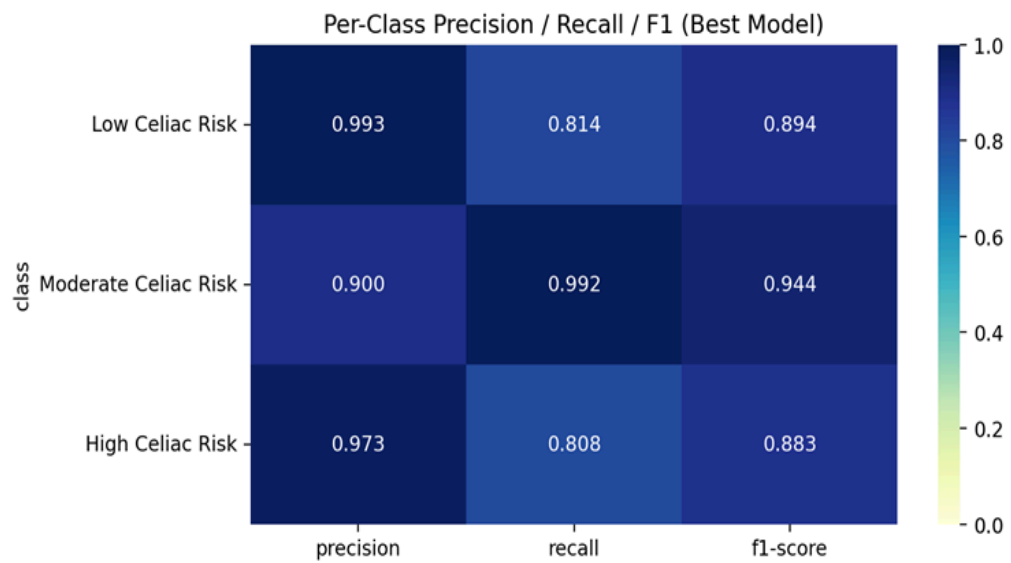
## 4.5 Model Report Charts and What They Show



## 4.1 Model Strategy Metric Comparison

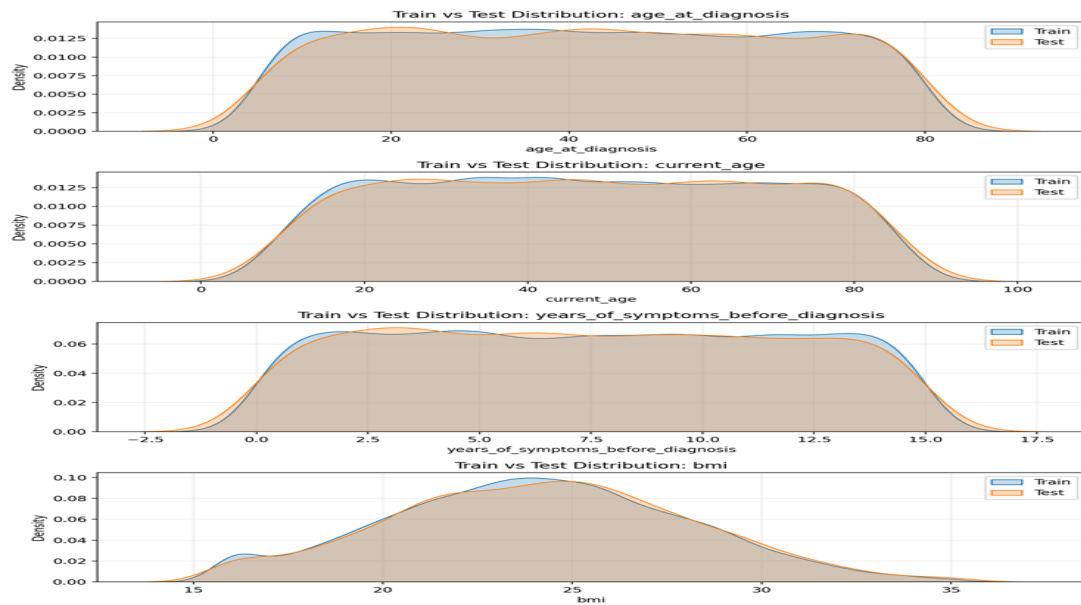


4.2 Best Model Radar Profile



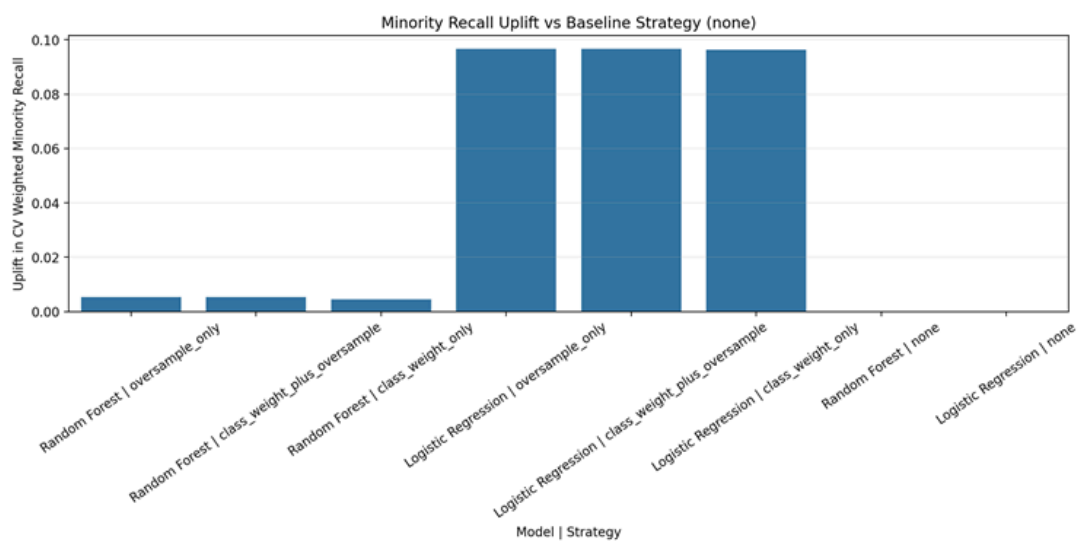
4.3 Per-Class Metrics Heatmap

This chart is important because implementation decisions optimize minority behavior. Per-class view confirms whether performance distribution is balanced enough for project goals.



#### 4.4 Train/Test Numeric Distribution Drift

This chart is relevant to preprocessing stability and split consistency. Major drift would indicate generalization risk or data-source inconsistency.



#### 4.5 Strategy Uplift vs Baseline

This chart is directly aligned with implemented strategy comparison logic and communicates practical value added by balancing and weighting decisions.

## 4.6 Implementation Code

```
import os

import warnings

from pathlib import Path

import joblib

import matplotlib.pyplot as plt

import numpy as np

import pandas as pd

import seaborn as sns

from sklearn.compose import ColumnTransformer

from sklearn.ensemble import RandomForestClassifier

from sklearn.linear_model import LogisticRegression

from sklearn.base import clone

from sklearn.metrics import accuracy_score, balanced_accuracy_score,
classification_report, confusion_matrix, f1_score, recall_score

from sklearn.model_selection import StratifiedKFold

from sklearn.pipeline import Pipeline

from sklearn.preprocessing import OneHotEncoder, StandardScaler

RANDOM_STATE = 42

np.random.seed(RANDOM_STATE)

DEFAULT_HF_MAIN_DATASET = "ayush-yadavv/nhanes_celiac_data"

HF_MAIN_DATASET = os.getenv("HF_MAIN_DATASET_SECRET") or
os.getenv("HF_MAIN_DATASET")

if not HF_MAIN_DATASET:

    try:

        from google.colab import userdata # type: ignore

        HF_MAIN_DATASET = userdata.get("HF_MAIN_DATASET_SECRET") or
DEFAULT_HF_MAIN_DATASET

    except Exception:

        HF_MAIN_DATASET = DEFAULT_HF_MAIN_DATASET

HF_TRAIN_SPLIT = "train"
```

```

HF_TEST_SPLIT = "test"

HF_MAIN_MAX_ROWS = None

# Imbalance handling strategies to compare during training.

BALANCING_STRATEGIES = [

    {"name": "none", "class_weight": False, "oversample": False},

    {"name": "class_weight_only", "class_weight": True, "oversample": False},

    {"name": "oversample_only", "class_weight": False, "oversample": True},

    {"name": "class_weight_plus_oversample", "class_weight": True, "oversample":
True},
]

# Resolve paths correctly whether the notebook kernel starts in the repo root or
train/.

CWD = Path.cwd().resolve()

if (CWD / "train").exists() and (CWD / "models").exists():

    PROJECT_ROOT = CWD

elif CWD.name == "train" and (CWD.parent / "models").exists():

    PROJECT_ROOT = CWD.parent

else:

    PROJECT_ROOT = CWD

# Keep model output compatible with backend defaults.

MODEL_OUTPUT_DIR = os.getenv("MODEL_OUTPUT_DIR")

OUTPUT_DIR = Path(MODEL_OUTPUT_DIR).expanduser() if MODEL_OUTPUT_DIR else
PROJECT_ROOT / "models"

OUTPUT_MODEL_PATH = OUTPUT_DIR / "celiac_risk_model.pkl"

OUTPUT_METADATA_PATH = OUTPUT_DIR / "model_metadata.pkl"

CV_FOLDS = 5

warnings.filterwarnings("ignore", message=".*IProgress not found.*")

NUMERIC_FEATURES = [

    "age_at_diagnosis",

    "current_age",

```



```

    "years_of_symptoms_before_diagnosis",

    "bmi",

    "followup_years",
]

CATEGORICAL_FEATURES = [

    "sex",

    "marsh_grade_at_diagnosis",

    "mucosal_healing_on_followup",

    "rcd_type",

    "smoking_status",

    "gfd_adherence",

    "family_history_of_malignancy",

    "hla_risk",
]

MISSING_INDICATOR_FEATURES = [f"{col}_was_missing" for col in NUMERIC_FEATURES]

MODEL_NUMERIC_FEATURES = NUMERIC_FEATURES + MISSING_INDICATOR_FEATURES

TARGET_COLUMN = "celiac_serology_risk"

EXPECTED_COLUMNS = MODEL_NUMERIC_FEATURES + CATEGORICAL_FEATURES + [TARGET_COLUMN]

VALID_CATEGORIES = {

    "sex": ["Male", "Female"],

    "marsh_grade_at_diagnosis": ["0", "1", "2", "3a", "3b", "3c"],

    "mucosal_healing_on_followup": ["Yes", "No"],

    "rcd_type": ["None", "RCD_I", "RCD_II"],

    "smoking_status": ["Never", "Former", "Current"],

    "gfd_adherence": ["Poor", "Partial", "Good", "Excellent"],

    "family_history_of_malignancy": ["Yes", "No"],

    "hla_risk": ["Low", "Medium", "High"],
}

NUMERIC_RANGES = {

    "age_at_diagnosis": (5, 80),

```

```

        "current_age": (5, 90),

        "years_of_symptoms_before_diagnosis": (0, 15),

        "bmi": (16, 35),

        "followup_years": (0, 20),

    )

def load_main_hf_dataset(dataset_name: str, split: str, max_rows: int | None =
None) -> tuple[pd.DataFrame, str]:

    from datasets import load_dataset

    split_expr = split if max_rows is None else f"{split}[:{max_rows}]"

    df = load_dataset(dataset_name, split=split_expr).to_pandas()

    source = f"hf:{dataset_name}/{split_expr}"

    return df, source

def fit_preprocessing_state(train_df: pd.DataFrame) -> dict:

    required_columns = NUMERIC_FEATURES + CATEGORICAL_FEATURES + [TARGET_COLUMN]

    missing_cols = [c for c in required_columns if c not in train_df.columns]

    if missing_cols:

        raise ValueError(f"Missing required columns in train split:
{missing_cols}")

    state = {"numeric_medians": {}, "categorical_fallbacks": {}}

    for col in NUMERIC_FEATURES:

        low, high = NUMERIC_RANGES[col]

        series = pd.to_numeric(train_df[col], errors="coerce")

        median_value = series.median()

        if pd.isna(median_value):

            median_value = (low + high) / 2

        state["numeric_medians"][col] = float(median_value)

    for col in CATEGORICAL_FEATURES:

        allowed = VALID_CATEGORIES[col]

        series = train_df[col].astype(str).str.strip()

```

```

        mode_value = series[series.isin(allowed)].mode(dropna=True)

        fallback = mode_value.iloc[0] if not mode_value.empty else allowed[0]

        state["categorical_fallbacks"][col] = fallback

    return state

def build_lr_pipeline(use_class_weight: bool) -> Pipeline:

    class_weight_option = "balanced" if use_class_weight else None

    return Pipeline(

        steps=[

            ("preprocessor", preprocessor),

            (

                "classifier",

                LogisticRegression(

                    max_iter=1000,

                    random_state=RANDOM_STATE,

                    class_weight=class_weight_option,

                ),

            ),

        ]

    )

def build_rf_pipeline(use_class_weight: bool) -> Pipeline:

    class_weight_option = "balanced" if use_class_weight else None

    return Pipeline(

        steps=[

            ("preprocessor", preprocessor),

            (

                "classifier",

                RandomForestClassifier(

                    n_estimators=200,

                    max_depth=10,

                    random_state=RANDOM_STATE,

```

```

        n_jobs=-1,

        class_weight=class_weight_option,

    ),

),

]

)

def cross_validate_strategy(

    pipeline: Pipeline,

    X_train: pd.DataFrame,

    y_train: pd.Series,

    use_oversampling: bool,

) -> dict:

    skf = StratifiedKFold(n_splits=CV_FOLDS, shuffle=True,
random_state=RANDOM_STATE)

    acc_scores, bal_scores, macro_f1_scores, minority_scores = [], [], [], []

    all_classes = np.array(sorted(y_train.unique()), dtype=int)

    class_to_pos = {int(c): i for i, c in enumerate(all_classes)}

    oof_proba = np.zeros((len(y_train), len(all_classes)), dtype=float)

    oof_true = y_train.values.astype(int)

    for fold_idx, (tr_idx, va_idx) in enumerate(skf.split(X_train, y_train),
start=1):

        X_tr = X_train.iloc[tr_idx].copy()

        y_tr = y_train.iloc[tr_idx].copy()

        X_va = X_train.iloc[va_idx].copy()

        y_va = y_train.iloc[va_idx].copy()

        if use_oversampling:

            X_tr, y_tr = oversample_training_data(X_tr, y_tr,
random_state=RANDOM_STATE + fold_idx)

```

```

fold_model = clone(pipeline)

fold_model.fit(X_tr, y_tr)

y_va_pred = fold_model.predict(X_va)

acc_scores.append(float(accuracy_score(y_va, y_va_pred)))

bal_scores.append(float(balanced_accuracy_score(y_va, y_va_pred)))

macro_f1_scores.append(float(f1_score(y_va, y_va_pred, average="macro",
zero_division=0)))

minority_scores.append(float(minority_recall_mean(y_va.values, y_va_pred)))

if hasattr(fold_model, "predict_proba"):

    fold_proba = fold_model.predict_proba(X_va)

    fold_classes = fold_model.classes_.astype(int)

    tmp = np.zeros((len(va_idx), len(all_classes)), dtype=float)

    for j, cls in enumerate(fold_classes):

        tmp[:, class_to_pos[int(cls)]] = fold_proba[:, j]

    oof_proba[va_idx] = tmp

    decision_tuning = tune_decision_weights_from_oof(oof_true, oof_proba,
all_classes)

return {

    "cv_accuracy": float(np.mean(acc_scores)),

    "cv_balanced_accuracy": float(np.mean(bal_scores)),

    "cv_macro_f1": float(np.mean(macro_f1_scores)),

    "cv_minority_recall_mean": float(np.mean(minority_scores)),

    "decision_weights": decision_tuning["weights"],

    "cv_weighted_macro_f1": decision_tuning["macro_f1"],

    "cv_weighted_balanced_accuracy": decision_tuning["balanced_accuracy"],

    "cv_weighted_minority_recall_mean":

```

```

decision_tuning["minority_recall_mean"],

    }

print(f"Selected model: {best['name']} | strategy={best['strategy']}")

print(f"Selected decision weights: {best['decision_weights']}")

print(f"Selected      CV      weighted      minority      recall      mean      (Low/High):
(best['cv_weighted_minority_recall_mean']:.4f)")

print(f"Selected CV weighted macro F1: {best['cv_weighted_macro_f1']:.4f}")

print(f"Selected      CV      weighted      balanced      accuracy:
(best['cv_weighted_balanced_accuracy']:.4f)")

print(f"Selected holdout macro F1: {best['macro_f1']:.4f}")

print(f"Selected holdout balanced accuracy: {best['balanced_accuracy']:.4f}")

print(f"Selected holdout accuracy: {best['accuracy']:.4f}")

OUTPUT_DIR.mkdir(parents=True, exist_ok=True)

strategy_lookup = {s['name']: s for s in BALANCING_STRATEGIES}

selected_strategy = strategy_lookup[best['strategy']]

metadata = {

    "numeric_features": ACTIVE_MODEL_NUMERIC_FEATURES,

    "base_numeric_features": NUMERIC_FEATURES,

    "missing_indicator_features": MISSING_INDICATOR_FEATURES,

    "active_missing_indicators": ACTIVE_MISSING_INDICATORS,

    "categorical_features": CATEGORICAL_FEATURES,

    "model_type": best['name'],

    "selection_strategy": best['strategy'],

    "decision_weights": best['decision_weights'],

    "accuracy": best['accuracy'],

    "balanced_accuracy": best['balanced_accuracy'],

    "macro_f1": best['macro_f1'],

    "minority_recall_mean": best['minority_recall_mean'],

```

```

    "cv_accuracy": best['cv_accuracy'],

    "cv_balanced_accuracy": best['cv_balanced_accuracy'],

    "cv_macro_f1": best['cv_macro_f1'],

    "cv_minority_recall_mean": best['cv_minority_recall_mean'],

    "cv_weighted_macro_f1": best['cv_weighted_macro_f1'],

    "cv_weighted_balanced_accuracy": best['cv_weighted_balanced_accuracy'],

    "cv_weighted_minority_recall_mean": best['cv_weighted_minority_recall_mean'],

    "cv_folds": CV_FOLDS,

    "data_source": data_source,

    "target_mode": train_cleaning_report['target_mode'],

    "target_column": TARGET_COLUMN,

    "target_type": "celiac_serology_risk",

    "not_malignancy_model": True,

    "intended_use": "Research prototype for estimating celiac risk from the
configured Hugging Face dataset.",

    "limitations": "Research prototype only; not for clinical diagnosis or
treatment decisions.",

    "effective_feature_note": "Feature coverage and quality depend on the
configured Hugging Face dataset schema.",

    "class_weight_balancing": selected_strategy['class_weight'],

    "random_oversampling": selected_strategy['oversample'],
}

joblib.dump(best_model, OUTPUT_MODEL_PATH)

joblib.dump(metadata, OUTPUT_METADATA_PATH)

print(f"Saved model: {OUTPUT_MODEL_PATH.resolve()}")

print(f"Saved metadata: {OUTPUT_METADATA_PATH.resolve()}")

sample = X_test.iloc[[0]].copy()

sample_truth = int(y_test.iloc[0])

```

```

sample_pred = int(best_model.predict(sample)[0])

sample_proba = best_model.predict_proba(sample)[0]

risk_labels = {0: "Low Celiac Risk", 1: "Moderate Celiac Risk", 2: "High Celiac Risk"}

print("Smoke test prediction:")

print(f"Ground truth: {sample_truth} ({risk_labels[sample_truth]}")

```

## 4.7 Summary

CeliGuard ML is implemented as a practical full-stack ML prototype with clear module boundaries and explicit contracts. The train module builds and evaluates models under a minority-sensitive strategy framework, then exports artifacts. The backend module loads those artifacts at startup and exposes stable prediction APIs. The frontend module validates backend readiness, collects schema-aligned user inputs, and renders interpretable outputs.

The implementation is technically coherent: feature contracts are aligned, runtime startup dependencies are managed, and artifact handoff between training and serving is explicit. The included report charts in `models/report_charts` provide additional evidence for strategy selection and model behavior analysis. At the same time, the current system remains prototype-grade and needs production hardening in security, monitoring, and automated validation.

From an engineering perspective, the codebase is a solid baseline for controlled iteration. The shortest path to higher maturity is not architectural replacement, but disciplined enhancement of testing, security controls, model governance, and runbook-based operations while preserving the existing train-serve-frontend structure.

The implementation is strong for a research prototype and already contains several good engineering patterns (schema validation, startup lifecycle controls, orchestration health checks, class-aware optimization). The main gap is not architecture but production hardening: security controls, automated tests, governance metadata, and operational observability.



## **CHAPTER 5**

### **RESULT ANALYSIS**

#### **5.1 Overview of Evaluation Approach**

This chapter discusses a performance evaluation of the current production version of the CeliGuard ML model using an artifact that has been created in conjunction with other repositories:

- models/model\_metadata.pkl
- models/report\_charts/\*
- Current Model - Random Forest (class\_weight\_only) - Target - celiac\_serology\_risk (low/moderate/high),
- Decision Weights - {0: 1.3; 1: 1.0; 2: 1.2}

The evaluation will be performed on metrics derived from hold-out sets and/or metrics associated with stratified 5-way CV metrics saved during the training of the model. As this is a risk stratification task with 3 classes, there will be a variety of measures (i.e., balanced accuracies, macro-F1 scores, minority recall means, etc.) considered in determining whether the performance has been concentrated in only one of the classes.

#### **5.2 Core Performance Metrics**

The accuracy of the model is high as it is shown to indicate an overall tendency for correct predictions. The model is predicted to make a correct prediction on approximately 93 of the 100 test samples in the holdout data set, while it is 93.82% accurate using 10-fold cross-validation. These values can be used as useful baseline signals for making predictive judgments about holdout data; however, when measuring total accuracy, it is important to use other metrics (e.g., not only holdout accuracy or cross-validation accuracy, but also some other metric) that are more sensitive to class membership.

Another important indicator of model performance is balanced accuracy, which is defined as the average of all class-level recall values. It is used to combine all classes into one measure and is more class-independent compared to regular accuracy; thus, high-balance accuracy values (0.87-0.89) indicate that the model is optimizing recall for all classes as well as for each individual risk class.

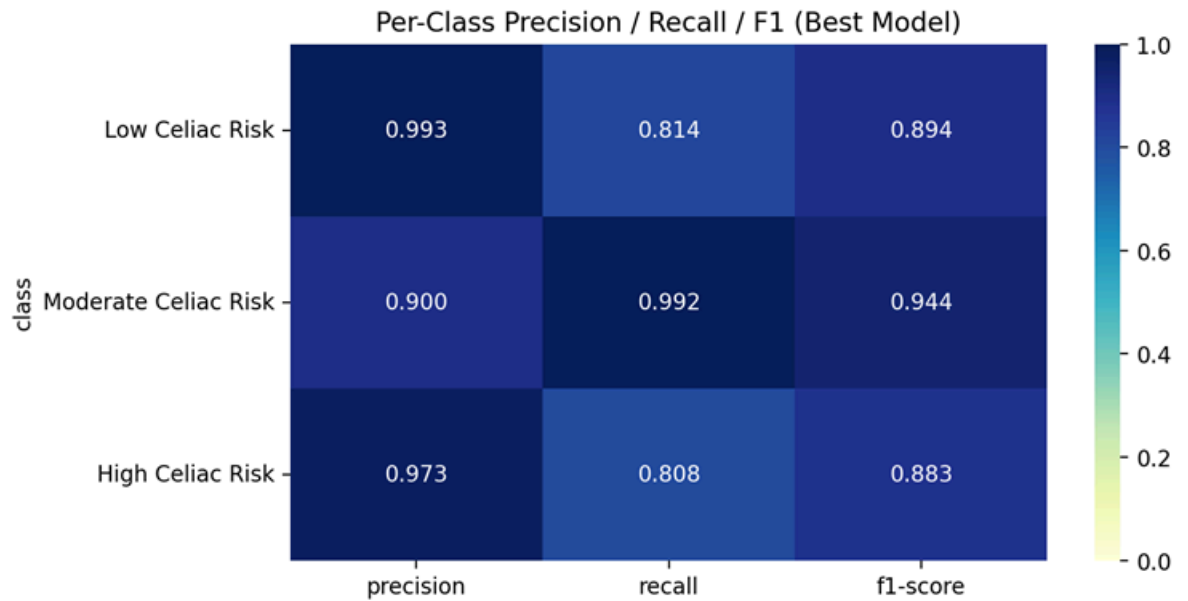
Macro-F1 is another means of evaluating the model's performance. Macro-F1 values greater than 0.90 indicate that models produce stable multiclass quality, with minimal variability in holdout compared to CV macro-F1 scores.

Minority recall is a key performance objective of the model; therefore, the training pipeline employs decision-weighting techniques as well as focuses on developing value-added enhancements to improve overall performance for the minority class. In the final model selection process, decision-weight adjustments were successful in producing an improvement in overall minority and boundary-risk class detection quality.

### **5.3 Strategy-Level Comparative Analysis**

The training workflow involved testing multiple different ways of balancing class distributions (i.e. no balancing, class weighting, oversampling, and combining these methods) in order to evaluate which were the most effective for the developed model. Final decisions regarding which method of balancing would be implemented in the model were based on an objective function ranking using the weighted mean of recall for minority classes as the primary ranking factor (with macro quality metrics being used for tie-breaking). This was an important factor in determining the appropriate method of balancing because a model can have higher overall accuracy scores than another model, but not be a more effective model if it is under-performing on the minority class that has been determined as risk-critical.

In addition to determining how to rank which of the balancing strategies would be used, the training workflow utilized class decision weights (i.e. {0:1.3, 1: 1.0, 2: 1.2}) in order to calibrate predictions after generating probabilities. Improvement of class-sensitive behavior and maintaining an acceptable level of macro-quality has occurred due to the use of class decision weights.



### 5.1 Confusion Matrix

The class-wise prediction behavior and error concentration can be assessed through the use of the confusion matrix visualisation. Assessment of these two factors is critical because the overall metrics alone provide very little insight into the reason(s) for a model's misclassification. Summary of the confusion matrix allows for verification of whether or not errors are concentrated in specific risk transitions and whether or not class-level performance is equal enough for use in risk stratification in practice.

### 5.4 Interpretation of Results

The deployed model exhibits measurable characteristics of performance in the following manner:

1. Excellent overall classification performance ( $> 92\%$  holdout accuracy).
2. Overall balanced performance of the classes (balanced accuracy  $\sim 0.8720$  holdout).
3. Excellent precision/recall balance within the individual classes (macro-F1  $\sim 0.9093$  holdout).
4. Acceptable sensitivity across the minority class (mean minority recall  $\sim 0.8102$  holdout).

The chosen Random Forest with class weighting is confirmed by both metric and visual diagnostics. The CV-to-holdout gap is controlled across primary metrics, indicating that the selection process has not resulted in overfitting solely to the cross-validation behaviour. The decision-weighted evaluation framework supports project intent by maintaining high-quality detection of minority classes while producing minimal macro-level degradation in the model output.

From a practical implementation perspective, there is consistency between the metrics evaluated and the pipeline design choices made during development:

a preference for evaluating multiclass metrics rather than utilizing accuracy-only rankings,

preference for selecting strategies that favour minority-aware utilities,

visual diagnostic tools were used to support interpretation of the model for review.

This enables result analysis to be reproducible and auditable based on only the repository artifacts.

## **CHAPTER 6**

### **CONCLUSION, LIMITATION AND FUTURE SCOPE**

#### **6.1 Conclusion**

The celiac risk stratification machine learning solution has a modular architecture with three primary components: training, backend inference, and frontend interaction. Data preprocessing, model comparison, evaluation of a balancing strategy, and export of metadata are all performed in the training workflow. The backend loads serialized artifacts and provides access to prediction and model status through API endpoints. The frontend takes structured data input, calls backend inferences, and displays risk classification, expected probabilities, and rationale. The orchestration of containers produces environments that allow for reproducible execution both locally and in the manner of deployment.

The model selected from the currently saved metadata (models/model\_metadata.pkl) was a Random Forest classifier utilizing the `class_weight_only` strategy, with decision weights of `{0: 1.3, 1: 1.0, 2: 1.2}`. The project produced recorded metrics that indicate strong predictive quality for the current configuration, including high overall accuracy, a very balanced result on a macro level, and reasonable recall for the minority class. These observations reflect the project's design decision to avoid optimizing globally and instead incorporate class-sensitive metrics in the selection of the model.

A key benefit of implementing these components is their uniformity; the feature schema used in training corresponds to backend validation and frontend payload creation, minimizing the risk of contract drift between these. The structure of the API response schema (`risk_class`, `risk_score` and `message`) is therefore compact and stable, making frontend integration and testing more straightforward. The health check for the startup provides further operational transparency by revealing the state of the loaded model.

The interpretation whitespace at the current metadata boundary will always be explicit, the `target_column` for this model is `'celiac_serology_risk'` and

not\_malignancy\_model='True'. Thus, all results should be interpreted as an assessment of the risk class for celiac disease within the context of a research prototype and NOT as a validated clinical diagnosis for malignancy. Within the boundary of this whitespace, the implementation is technically coherent, repeatable and expandable.

## **6.2 Limitations**

There are a number of limitations in the current implementation of the celiac serology risk predictor:

1. Defining Targets- The deployed target is celiac\_serology\_risk, which is not a valid malignant endpoint. Thus, oncologists will not be able to use these findings directly in the clinical context.
2. Data Sources- The performance characteristics of the model are sensitive to the quality of the input training data, the stability of the schema, and the class distribution of the training data used. If the input training data become unreliable due to upstream drift or inconsistency between the labels in the dataset, the reliability of the output prediction will be affected.
3. External Validation- The output currently being presented by this tool have not been validated externally across at least one independent external cohort.
4. Explainability- Explanations for the outputs currently being generated are based on rules, whereas true attribution methods (i.e., SHAP/LIME) must be used to fully develop the local explanation for a prediction.
5. Production Security- Many default settings for the prototype repository are permissive (i.e., CORS), and the prototype has not yet implemented endpoint authentication, authorization, or rate limiting built into the prototype.
6. Observability & Audit- The current version has limited capabilities for long-term audits of the predictors, limited structured monitoring of the models, and limited advanced diagnostics of the models while they are running in production.
7. Automated Quality Gates- The CI process for performing testing on the contract URLs, on the metadata invariants, and establishing thresholds for retraining due to

regression is not complete.

#### 8. Lifecycle Governance Maturity

Metadata is present; however, there are still many areas of the lifecycle workflow that are in progress including the Building of a Registry of Formal Models, Promotion Policy and Automating the Rollback Process.

#### 9. Absence of Temporal Modelling

The present inference is based on single time points; there is no longitudinal trajectory to model across multiple ante-grade Subsequent Observation Periods.

These limitations define the dividing line between what is considered a "Robust" Research Prototype and a "Production Grade" Clinical Support System.

### 6.3 Future Scope

The next steps for further development should adhere to a phased approach around the roadmap used for enhancing scientific validity, engineering reliability, and operational governance:

#### 1. Alignment of target(s) and labeling

If the program scales into more oncology-related end points, the definition of the targets and labeling strategy must be updated and aligned across the entire training code, metadata, API documentation, and user interface language.

#### 2. Dataset expansion/representation

More diverse demographic and clinical variety will increase generalizability and decrease the chance of bias being introduced. The new populations being added to the cohort(s) should include differences in variables such as age, geographic location and the profile of disease progression, etc.

#### 3. Improved validation processes

Independent external testing cohorts, as well as subgroup audits and calibration-based reporting processes should be added to determine reliability during distributional shifts.

#### 4. Improve explainability

Add SHAP/LIME workflows to provide local feature attribution reporting so that each individual prediction can now be more easily interpreted during a clinical review.

#### 5. Longitudinal modeling path

Extend from static snapshots to time-aware modeling that captures follow-up trajectories and different ways in which one evolves over time.

#### 6. Establish protocols for governing models

Include registry-style protocols for establishing models, version identification tags associated with each artifact, immutable run records, and gate controls for promoting and rolling back models.

#### 7. Implement security and access control measures

Move away from initial prototype permissions defaults to prohibited cross-origin resource sharing for restricted-access endpoints, use of authenticated actions, implementation of authorization policy controls, and abuse prevention measures.



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